

19 2026-03-06 | Week 08 | Lecture 19

19.1 Continuous random variables

(This lecture is based on sections 4.1-4.2 in the textbook)

Definition 81 (pdf). Let X be a continuous random variable. The **probability density function** (aka “pdf” or “density”) of X is a function f_X such that for any real numbers a, b with $a \leq b$, we have

$$\mathbb{P}[X \in (a, b)] = \int_a^b f_X(x) dx$$

The graph of f_X is called the **density curve** of X .

Observations about pdfs:

- $f_X(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- $f_X(x)$ is not a probability because it could be greater than 1. Instead, you can think of it vaguely as a “relative likelihood”.

Proposition 82. Let $F(x) = \mathbb{P}[X \leq x]$ be the cdf of X , and let f be the pdf of X . Then

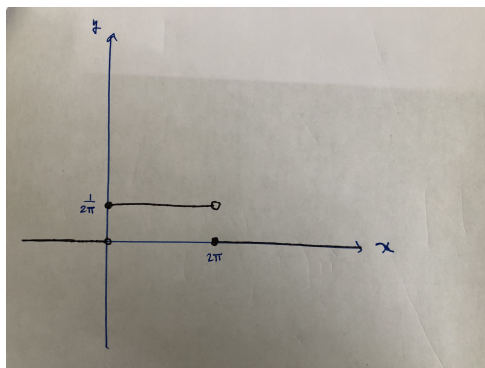
$$F(x) = \int_{-\infty}^x f(t) dt \quad \text{and} \quad F'(x) = f(x)$$

Proposition 83 (Properties of cdfs of uniform random variables).

- $\mathbb{P}[a \leq X \leq b] = F_X(b) - F_X(a)$
- F is a nondecreasing function
- $F(-\infty) = 0$ and $F(+\infty) = 1$.

Example 84 (Pinwheel example continued). Recall the example of the pinwheel Example 79, with the angle X measured in radians, so that the state space of X is $[0, 2\pi)$, with all values equally likely. Recall that we computed the pdf as

$$f_X(x) = \frac{1}{2\pi} \cdot \mathbf{1}_{[0, 2\pi)}(x) = \begin{cases} \frac{1}{2\pi} & : x \in [0, 2\pi) \\ 0 & : x \notin [0, 2\pi) \end{cases}$$



Recall also that the cdf is defined as

$$F_X(x) := \int_{-\infty}^x f_X(t) dt.$$

So that for the pinwheel angle X , we have

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x \frac{1}{2\pi} \cdot \mathbf{1}_{[0,2\pi)}(t) dt \\ &= \frac{1}{2\pi} \int_{-\infty}^x \mathbf{1}_{[0,2\pi)}(t) dt. \end{aligned} \tag{12}$$

To evaluate an integral of this form, we have to consider three cases:

- $x < 0$ (i.e. when x is to the left of the interval)
- $x \geq 2\pi$ (i.e. when x is to the right of the interval)
- $0 \leq x < 2\pi$ (i.e. when x is in the interval)

Let's think through these cases:

- When $x < 0$, the integral on the right-hand side of Eq. (12) is zero.
- When $x \geq 2\pi$, the integral equals

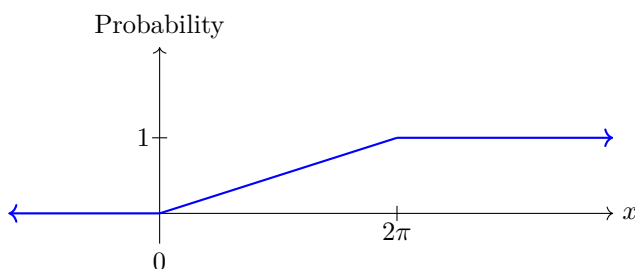
$$\int_{-\infty}^0 0 dt + \frac{1}{2\pi} \int_0^{2\pi} dt + \int_{2\pi}^x 0 dt = 0 + 1 + 0 = 1$$

- When $0 \leq x < 2\pi$, the integral equals

$$\int_{-\infty}^0 0 dt + \frac{1}{2\pi} \int_0^x dt = 0 + \frac{x}{2\pi} = \frac{x}{2\pi}$$

From this analysis and Eq. (12), we can conclude that

$$F_X(x) = \begin{cases} 0 & : x < 0 \\ \frac{x}{2\pi} & : 0 \leq x < 2\pi \\ 1 & : x \geq 2\pi \end{cases}$$



End of Example 84. $\square$

Definition 85 (Uniform random variable). Let I be an interval with endpoints a and b , such that $a < b$. A continuous random variable X has **uniform distribution** on I if it has pdf

$$f(x) = \begin{cases} \frac{1}{b-a} & : x \in I \\ 0 & : x \notin I \end{cases}$$

In this case, X is called a **uniform** random variable.

Definition 86 (Exponential random variable). We say that X is an **exponential** random variable with rate $\lambda > 0$ if it has pdf

$$f(x) = \lambda e^{-\lambda x} \mathbf{1}_{[x \geq 0]}.$$

Remark 87. In Definition 86, the term $\mathbf{1}_{[x \geq 0]}$ is an “indicator function” which takes value 1 when $x \geq 0$ and 0 if $x < 0$.

Example 88 (Waiting times). Suppose you station yourself at spot on a road and watch as cars pass by. Let T be the time in minutes until the first car passes by. This random variable is called the *waiting time*.

Under certain circumstances, it is reasonable for T to have pdf

$$f(x) = \lambda e^{-\lambda x} \mathbf{1}_{[x \geq 0]} \quad (13)$$

for a certain (fixed) value of $\lambda > 0$.

For concreteness, let's assume that $\lambda = 3$, so

$$f(x) = 3e^{-3x} \mathbf{1}_{[x \geq 0]}$$

Then the cdf is

$$\begin{aligned} F(t) &= \int_{-\infty}^t f_T(x) dx \\ &= \int_0^t 3e^{-3x} dx \\ &= 3 \left[\frac{-e^{-3x}}{3} \right]_0^t \\ &= 1 - e^{-3t}. \end{aligned}$$

Equivalently,

$$\mathbb{P}[T > t] = e^{-3t}.$$

For example the probability that you have to wait more than $t = 30$ seconds for a car to pass by is

$$\mathbb{P}[T > 5] = e^{-3 \cdot 0.5} \approx .22$$

What about

$$\mathbb{E}[T]?$$

If the cars come at a rate of $\lambda = 3$ cars per minute, then on average how long would you have to wait for a car to arrive? This should be 20 seconds.

End of Example 88. $\square$

Definition 89 (Expected value – continuous random variables). The **expected value** of a continuous random variable X is

$$\mathbb{E}[X] := \int_{-\infty}^{\infty} x f(x) dx$$

where $f(x)$ is the pdf of X . (Provided that the integral above converges absolutely).

We also have an analogue of the law of the unconscious statistician (Theorem 60)

Theorem 90 (LOTUS - continuous case). *Let X be a continuous random variable with pdf f . Then for any function $h : \mathbb{R} \rightarrow \mathbb{R}$,*

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx,$$

provided that the integral exists and is finite.

Suppose $\mu = \mathbb{E}[X]$. Applying Theorem 90 to the function $h(x) = (x - \mu)^2$ gives the following formula for the variance of a continuous random variable:

$$\text{Var}(X) := \mathbb{E}[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx.$$

And again, we have the shortcut formula

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$